

Manoel Da Ponte

AI ENGINEER • DATA ENGINEER

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AI and data engineer, co-founder of WiseTwin: LLM agents, RAG pipelines and a multitenant SaaS platform in production. French citizen (EU/EFTA national), based in Switzerland.

[For more details and project examples, visit manoel-daponte.com](#)

Skills

Generative AI & Agents	LLM agents (tool use), RAG, pgvector, LLM-as-judge, SSE, multimodal vision
Data Engineering	ETL, PostgreSQL, pgvector, SQLAlchemy, Prisma, pandas, Prefect
Software Development	Python, FastAPI, TypeScript, Next.js, React, Node.js
Cloud & DevOps	Azure, Docker, GitHub Actions, Vercel, AWS
Real-Time 3D	PlayCanvas, Gaussian Splatting, GLSL shaders, Unity
Languages	French (native), English (C1)

Work Experience

WiseTwin	Dunkirk, France (remote)
CO-FOUNDER & LEAD DEVELOPER	March 2024 - Present
- Multitenant SaaS platform for industrial safety training: 80 000 lines of TypeScript, 124 API routes, SSO, MFA, xAPI/Qualiopi exports - HSE AI copilot: LLM agent with 12 tools, pgvector RAG over 62 726 industrial accidents, SSE streaming, multimodal vision - Automated course generation from PowerPoint: Whisper transcription, LLM writing, quizzes verified by LLM-as-judge - Web-based 3D editors (Gaussian Splatting, Google Maps 3D); Azure Container Apps infrastructure, GitHub Actions CI/CD - Per-organization AI cost governance (budgets, quotas, token tracking); tenant isolation enforced by CI tests - Technologies: Python, FastAPI, TypeScript, Next.js, Claude, pgvector, PostgreSQL, Docker, Azure	
TotalEnergies	Esbjerg, Denmark
DATA ENGINEER	March 2022 - February 2024
- End-to-end data projects in small teams (1 to 3 people): data management, backend and frontend - Accident prevention (IoT, NLP, LLM fine-tuning), invoice control through computer vision, large-scale SAP extraction - Offshore IoT architecture: data warehouse, anomaly detection, Prefect monitoring (Python, Azure, SQL) - Application deployments on Azure and on-premise through CI/CD pipelines	
CGI	Toulouse, France
DATA SCIENTIST	July 2020 - March 2022
- Documentation chatbot with semantic search (Elasticsearch) and data migration on AWS (Python, NLP, SQL) - Behavioral analytics and personalized recommendations to improve user journeys	

Selected Projects

WiseAtlas	WiseTwin
DESIGNER & DEVELOPER	2025 - 2026
SaaS editor for photorealistic 3D maps (Google Maps 3D): geolocated GLB models, LLM-powered multilingual translation	
Splat Editor SafetyTour	WiseTwin
DESIGNER & DEVELOPER	2025 - 2026
Web-based 3D Gaussian Splat editor: industrial scene cleanup, GPU selection through GLSL shaders, hotspots with quizzes	

Education

MSc in Data Science and Engineering	Paul Sabatier University, Toulouse III
GRADUATE	2019 - 2021